Fast and Reliable TCP:
Sliding Window & Cumulative ACKs

CS698 Assignment
Fall 2025
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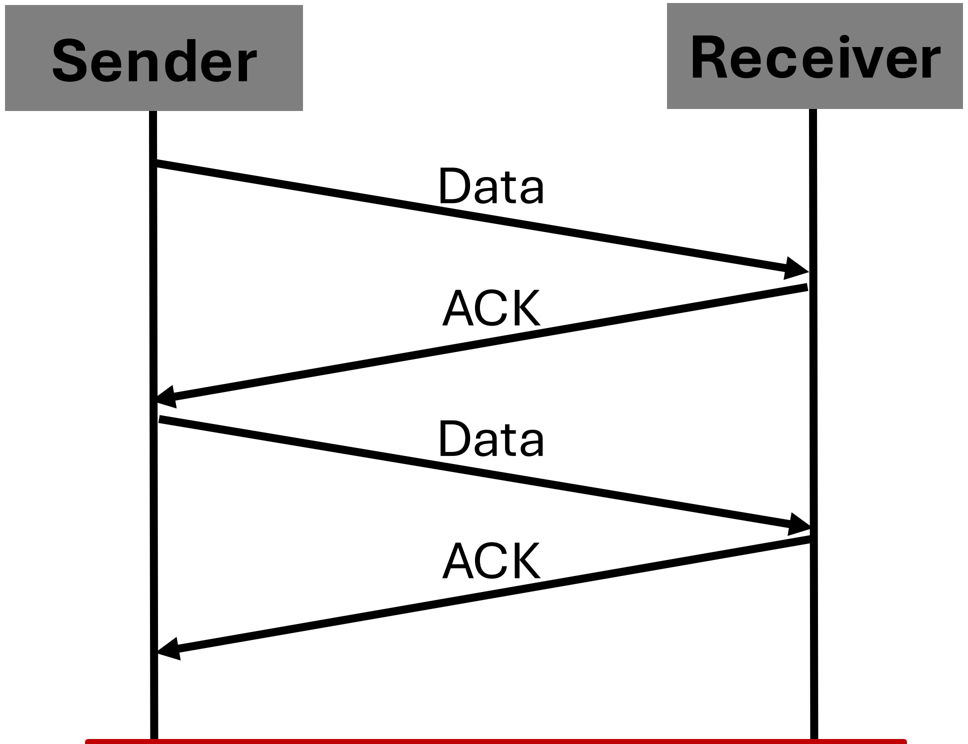
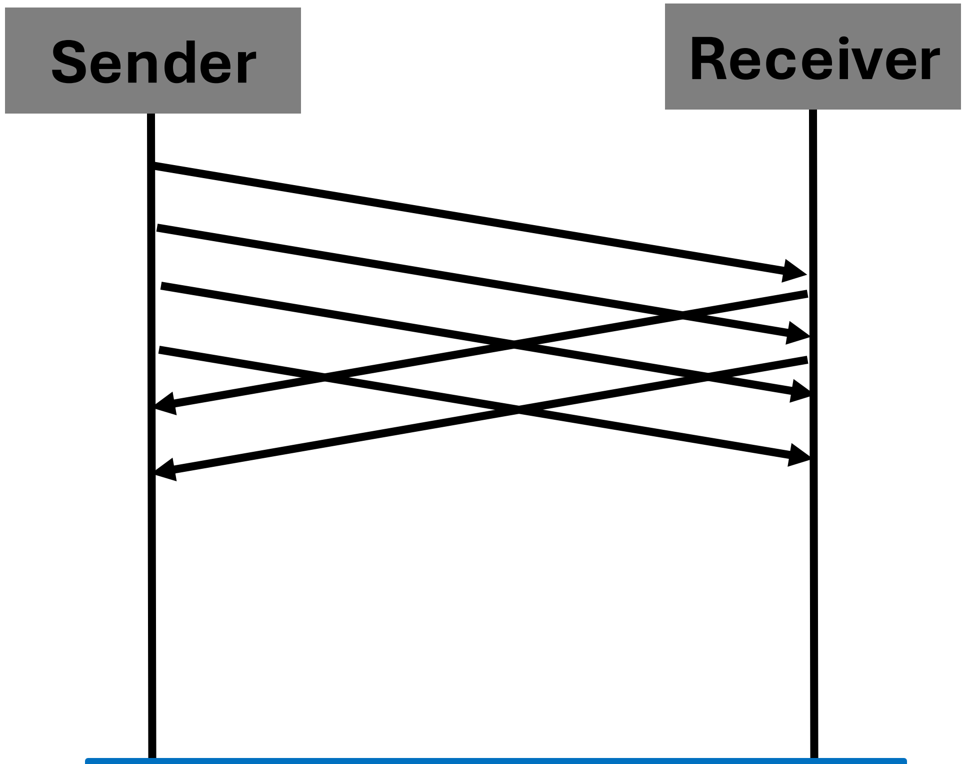
Motivation

- **Why Do We Need Sliding Windows?**



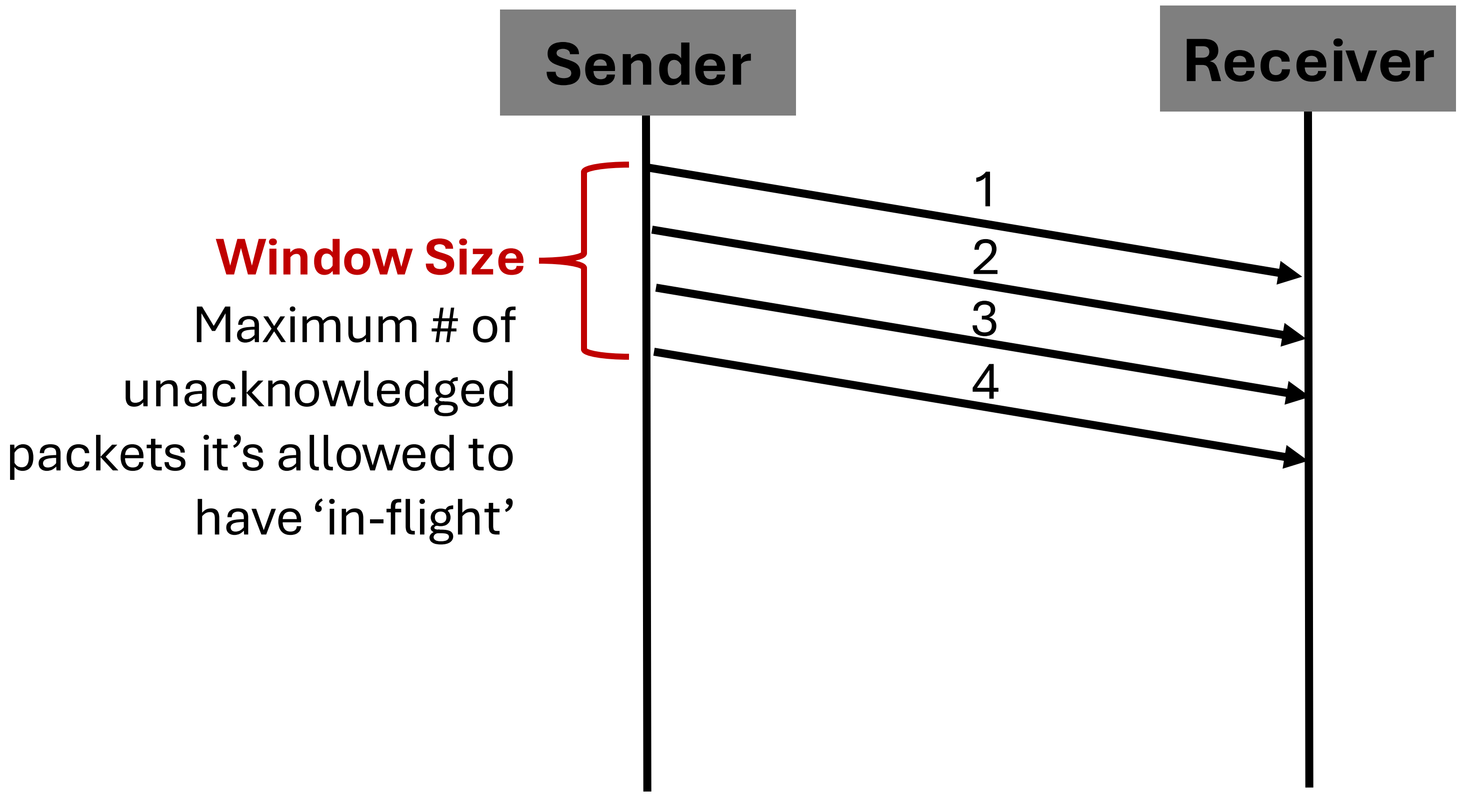
Slow and Inefficient



Pipelining

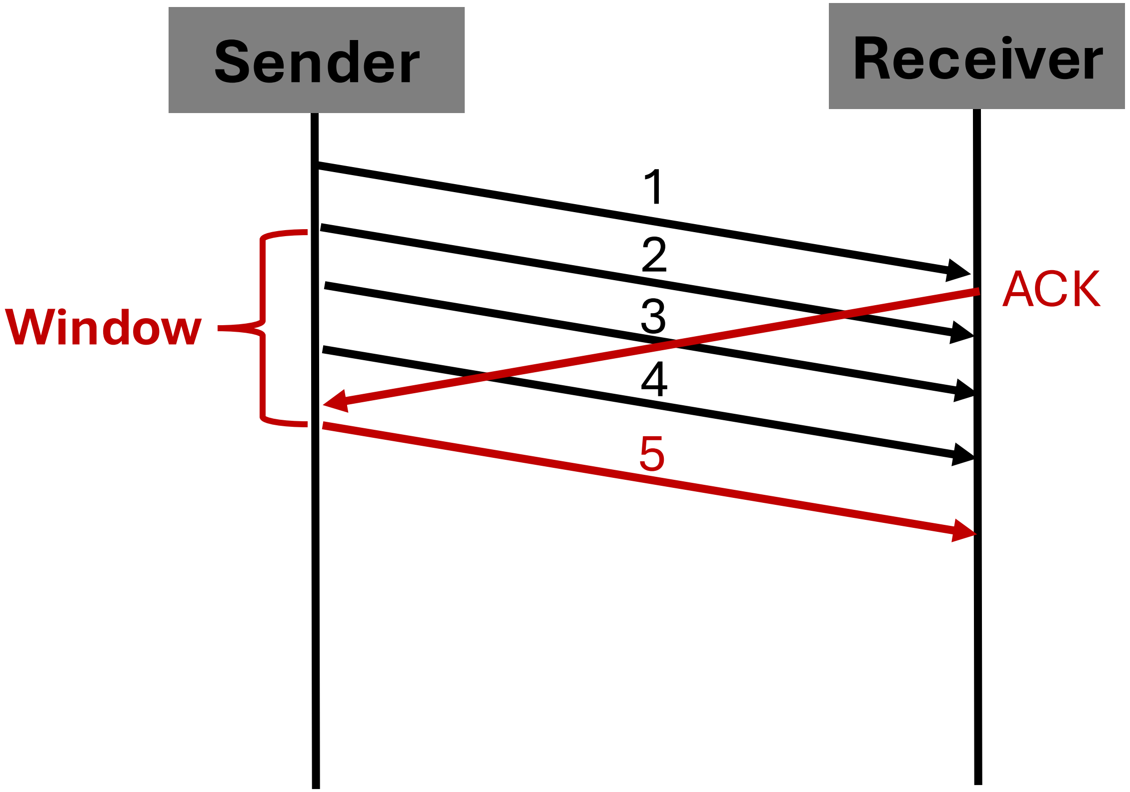
Sliding Window – Basic Idea

- **Send multiple packets before ACKs**



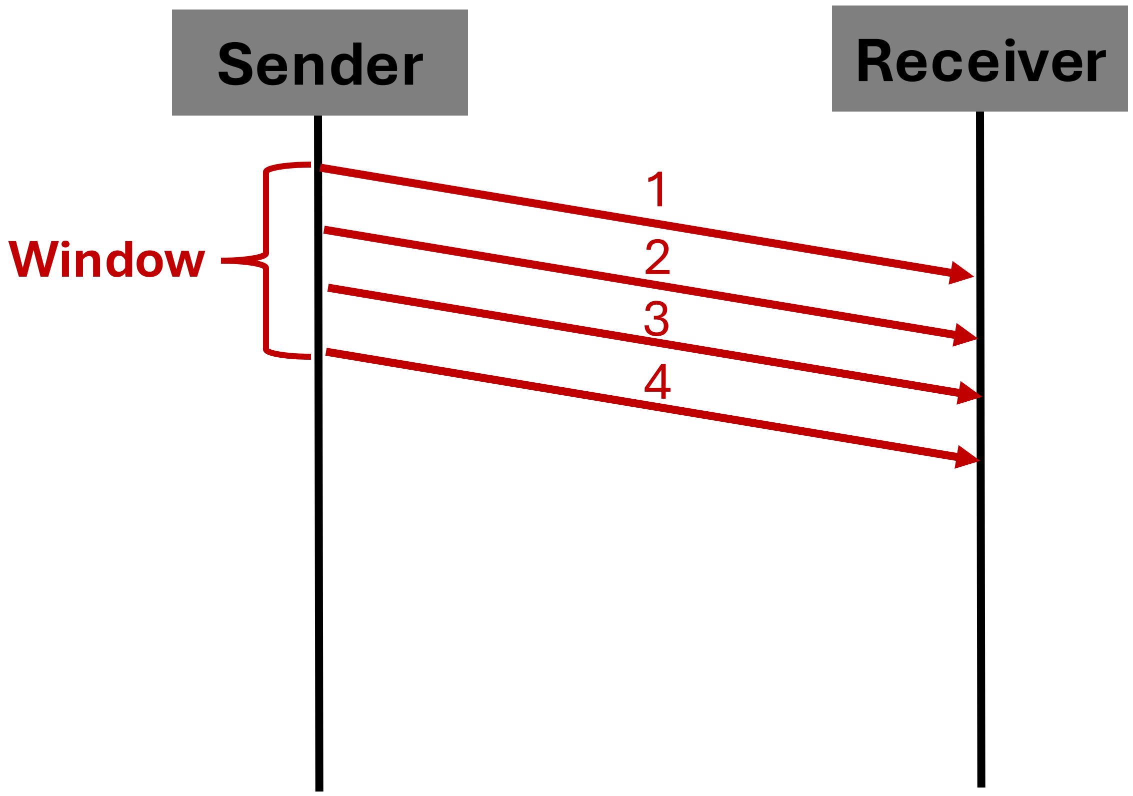
Sliding Window – Basic Idea

- Slides the window when ACK comes in



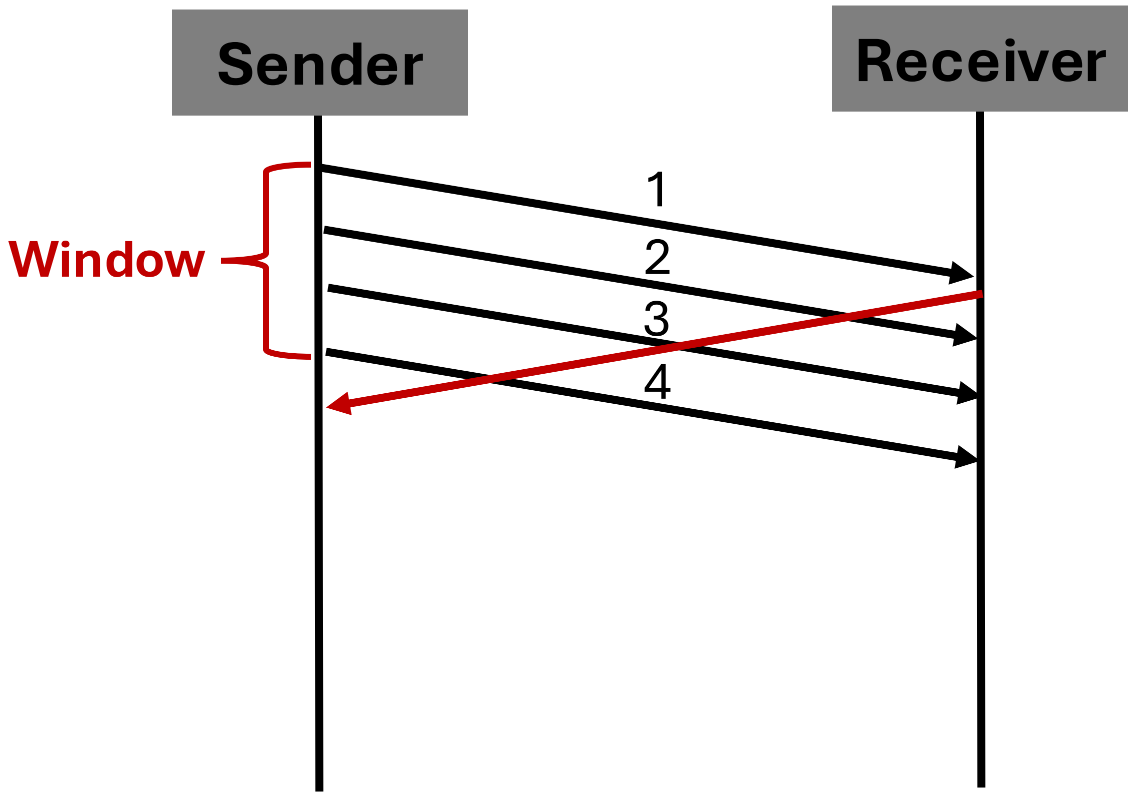
Sliding Window – Example

- An example with a window size of 4



Sliding Window – Example

- An example with a window size of 4



Sliding Window – Example

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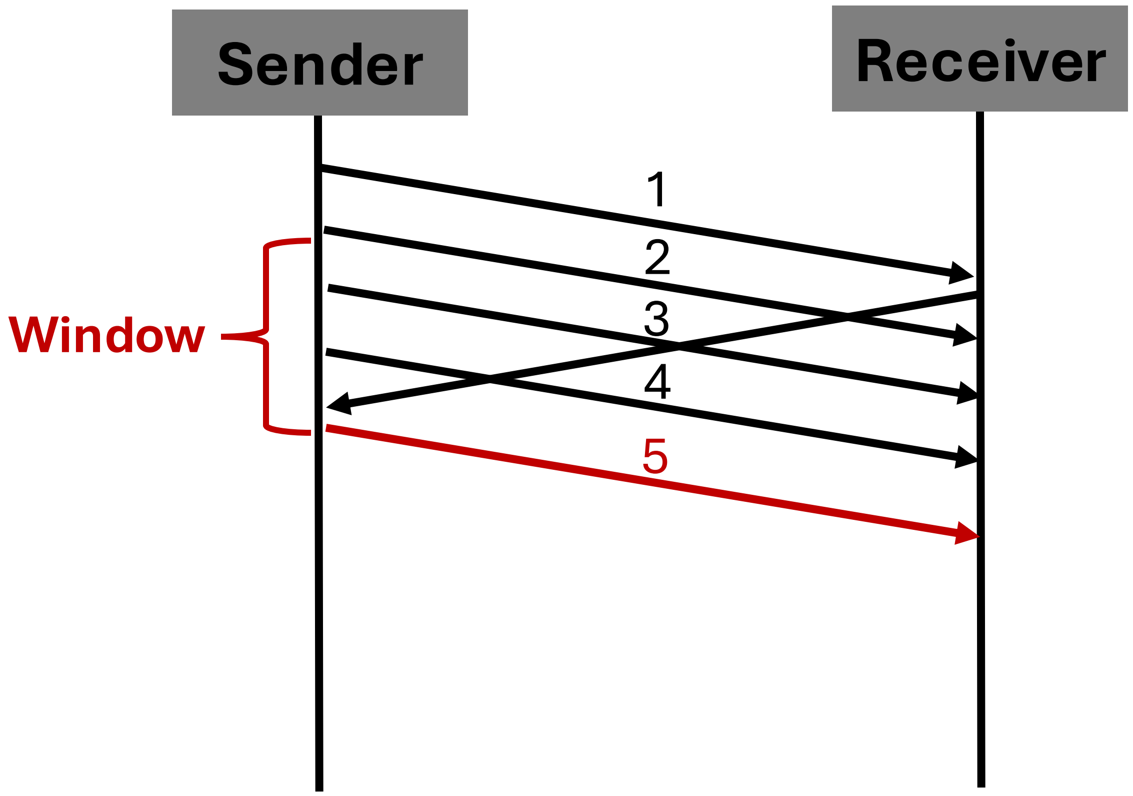
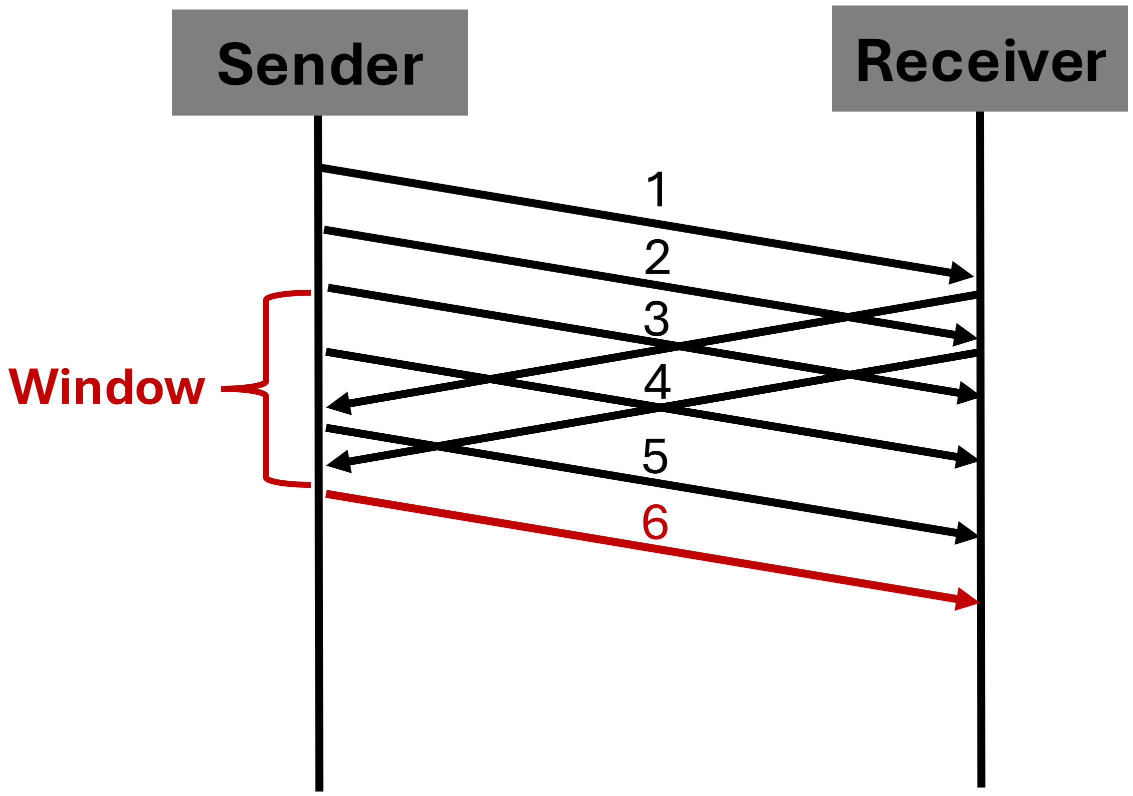


Sliding Window – Example

- An example with a window size of 4



Sliding Window – Example

- **Window size?**

Too small -> under utilization

Too large -> congestion and loss

TCP dynamically adjusts window size

Cumulative ACKs

- **What is a cumulative ACK?**

The ACK number represents the next expected packet

“ACK 5”

“I have received everything up to 4, and I’m now expecting 5!”

Cumulative ACKs

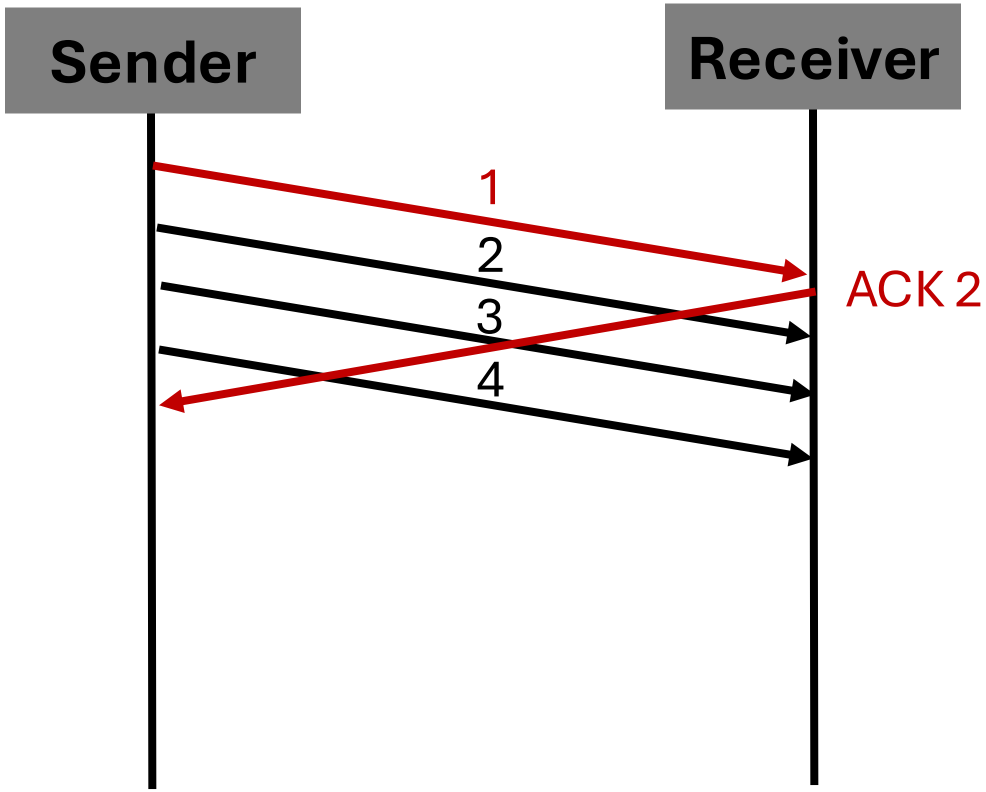
- **Why useful?**

Simpler for receiver

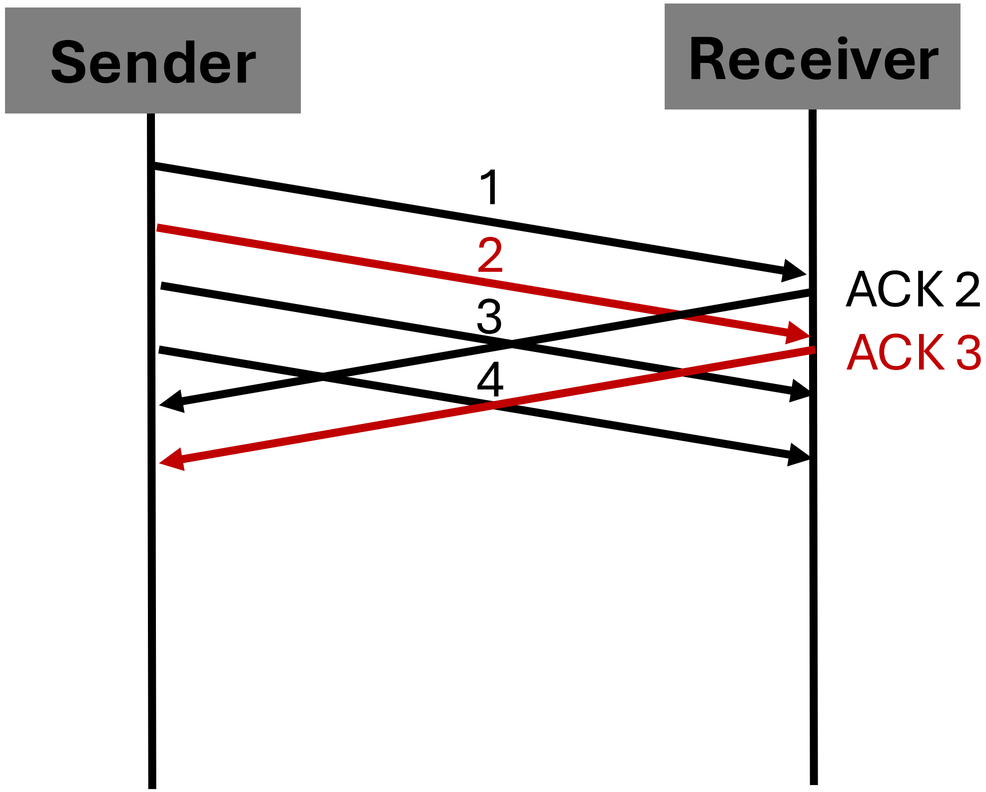
Lower overhead

More robust to packet reordering

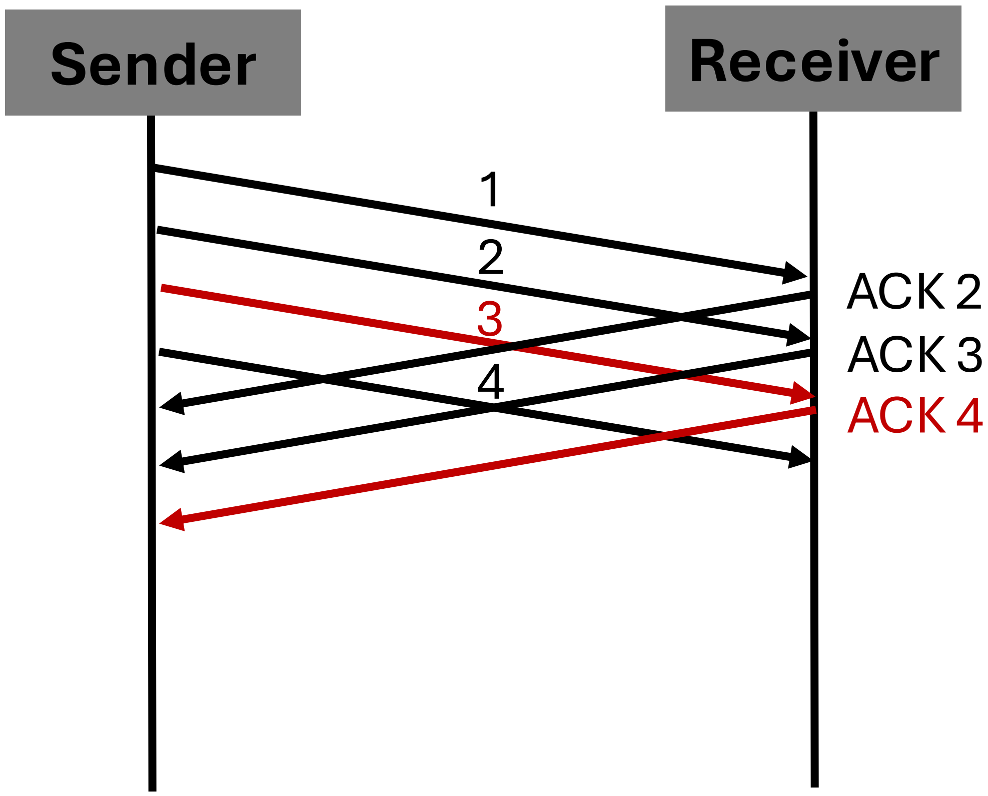
Cumulative ACK – Example



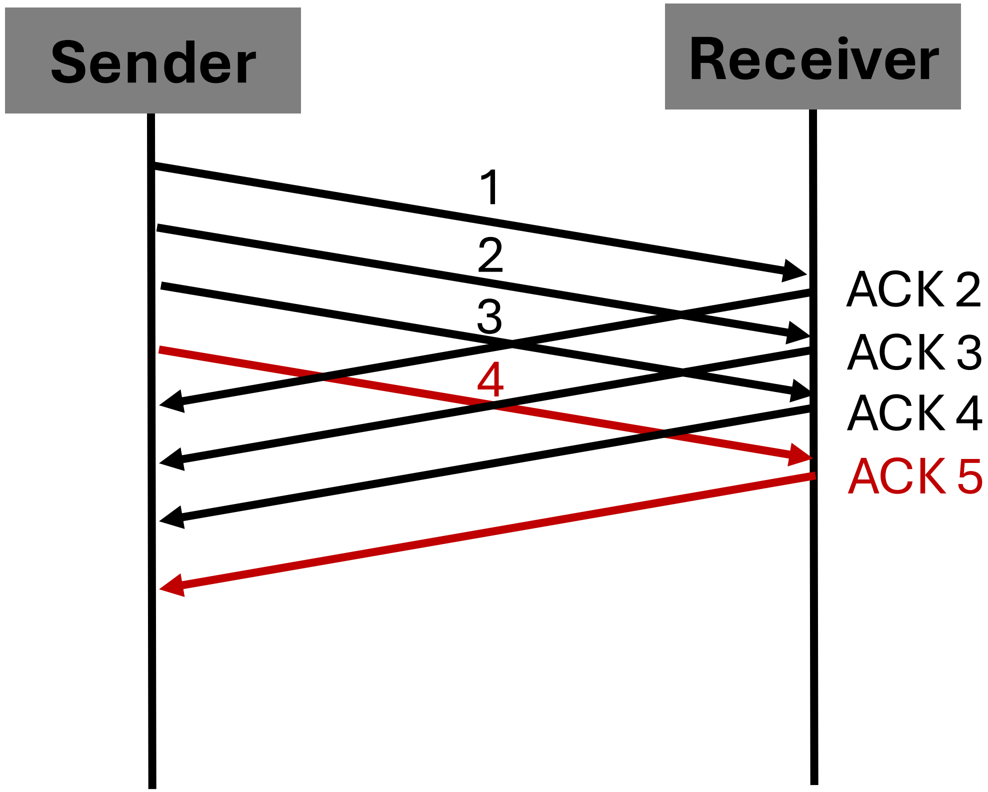
Cumulative ACK – Example



Cumulative ACK – Example

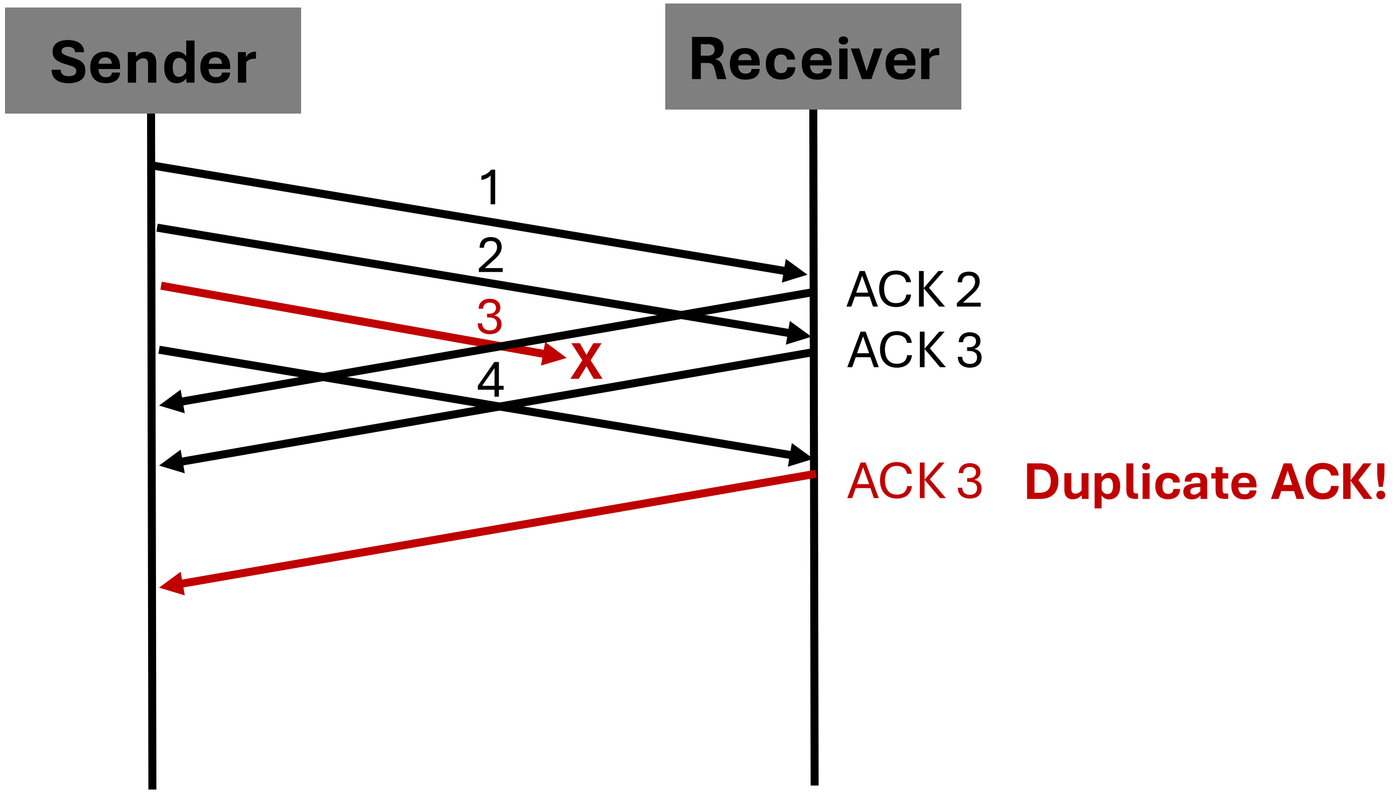


Cumulative ACK – Example



Cumulative ACK – Loss Handling

- **What if a packet is lost?**



Putting it all together

Sliding Window = Parallelism

Cumulative ACKs = Efficient Feedback

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TCP with

Reliable

High-throughput

Adaptive to Network

Summary

- Why sliding windows exist
- How the sender window shift
- How cumulative ACKs work
- How TCP handles loss using duplicate ACKs

Preview of Next Lecture

- **Flow Control and Congestion Control**

Window size is determined by:

Flow control – protects the receiver

Congestion control – protects the network

Flow Control

Receiver buffer is limited
Receiver advertises a receiver window
Sender ensures in-flight <= receiver window
Prevents receiver overflow

Congestion Control

Network routers also have limited buffers
TCP adjusts a window based on conditions
Grows when no loss; shrink when congestion is detected
Prevents congestion collapse

References

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Thank you!

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